

Two-Stage Feature Selection and Localization for Distribution-Shift Attribution in a Marketplace

Technical write-up

Abstract

We formalize a pipeline that, given a new batch of relational marketplace data, (i) *detects* whether the distribution has shifted, (ii) *localizes* the shift to the responsible features through a two-stage feature-selection procedure with hierarchical error control, along two entity axes (supply and demand), and (iii) turns the result into a structured, machine-consumable insight carrying a *retrain decision*. Detection uses an exact permutation Maximum Mean Discrepancy (MMD) test for covariate shift and a pseudo-outcome / R -risk test for concept drift. Localization combines leave-one-group-out (LOGO) and exact Shapley attribution over the group power set with a Westfall–Young step-down max-T permutation procedure (family-wise error control, robust to correlation) and subsampling stability selection. The decision layer estimates covariate and concept impact on a downstream loss via propensity-based importance weighting, gated by a batch-separability (AUC) overlap diagnostic. We give definitions, estimators, validity statements, algorithms, and the structured-insight schema.

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1 Problem setup and notation

A marketplace generates *orders* (observations). Each order o records a buyer (user) $u(o)$, an item $i(o)$, and its merchant $m(o)$ (via item $\rightarrow$ merchant), an order identifier, and raw numeric attributes $a(o) \in \mathbb{R}^A$ (e.g. GMV, quantity, discount, rating). We compare an *existing* batch (label $W = 0$) against a *new* batch ($W = 1$).

The shift may be examined along different *entity axes* $e \in \{\text{merchant, buyer, } \dots\}$. Fix an axis with key $k_e(o)$ and batch label W_e defined at the entity level. Aggregation rolls order-level attributes up to the entity dimension.

Definition 1 (Aggregation map). *For entity j let $O_j = \{o : k_e(o) = j\}$ be its orders. The feature map Φ produces $x_j = \Phi(O_j) \in \mathbb{R}^p$ by applying a fixed set of aggregators to each raw attribute: summary statistics $\{\text{mean, sd, min, max, median, } q_{25}, q_{75}, \text{sum}\}$, rolling-window statistics over the entity’s order stream ordered by order id, and activity counts. Feature names factor as **attr** – **agg**, inducing a grouping $g : \{1, \dots, p\} \rightarrow \mathcal{G}$ by raw attribute.*

Let P_0, P_1 be the distributions of x on the two batches, with samples $X = \{x_i\}_{i:W_i=0}$ (m points) and $Y = \{y_i\}_{i:W_i=1}$ (n points). The global null is $H_0 : P_0 = P_1$.

2 Detection

2.1 Maximum Mean Discrepancy

Let $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ be a characteristic kernel (we use the RBF kernel $k(x, x') = \exp(-\gamma\|x - x'\|^2)$ with γ from the median heuristic on the pooled sample) with RKHS $\mathcal{H}$ and mean embedding $\mu_P = \mathbb{E}_{x \sim P}[k(x, \cdot)]$.

Definition 2 (MMD). $\text{MMD}^2(P_0, P_1) = \|\mu_{P_0} - \mu_{P_1}\|_{\mathcal{H}}^2 = \mathbb{E}[k(x, x')] + \mathbb{E}[k(y, y')] - 2\mathbb{E}[k(x, y)]$, with $x, x' \sim P_0$, $y, y' \sim P_1$ independent. For characteristic k , $\text{MMD}^2(P_0, P_1) = 0 \iff P_0 = P_1$.

Definition 3 (Unbiased estimator).

$$\widehat{\text{MMD}}_u^2 = \frac{1}{m(m-1)} \sum_{i \neq j} k(x_i, x_j) + \frac{1}{n(n-1)} \sum_{i \neq j} k(y_i, y_j) - \frac{2}{mn} \sum_{i,j} k(x_i, y_j). \quad (1)$$

The diagonal is excluded so that $k(x_i, x_i) = 1$ does not bias the within-sample terms; this fact is central to the bootstrap caveat in Section 5.

2.2 Permutation test and validity

Given the pooled sample $Z = X \cup Y$ with labels, draw permutations π of the labels and recompute the statistic. The p -value is $\hat{p} = (1 + \sum_{b=1}^B \mathbb{1}\{T_b \geq T_{\text{obs}}\}) / (1 + B)$.

Proposition 1 (Exact validity). *Under H_0 the pooled points are exchangeable, so for any statistic T the permutation p -value satisfies $\mathbb{P}_{H_0}(\hat{p} \leq \alpha) \leq \alpha$ for all $\alpha \in (0, 1)$, exactly and without distributional assumptions.*

2.3 The k -sample case

For $\kappa > 2$ batches $P_1, \dots, P_\kappa$, testing $H_0 : P_1 = \dots = P_\kappa$ requires aggregating pairwise divergences. A natural omnibus (the kernel analogue of DISCO / energy-distance ANOVA) is

$$T_{\text{omni}} = \sum_{r < s} \frac{n_r n_s}{N} \widehat{\text{MMD}}_u^2(P_r, P_s), \quad N = \sum_r n_r, \quad (2)$$

with a κ -way label permutation null. A maximum variant $\max_{r < s} \widehat{\text{MMD}}_u^2$ targets a single divergent pair. The composite null means rejection must be followed by pairwise/subset localization (Section 3); this is exactly the two-stage logic.

2.4 Concept drift via the causal objective

Detection above concerns $P(X)$. To detect a change in the outcome mechanism $P(Y | X)$ we reuse the project’s causal-objective machinery. With batch as “treatment” W , outcome Y , nuisances $\mu(x) = \mathbb{E}[Y | x]$ and $e(x) = \mathbb{P}(W = 1 | x)$ estimated by cross-fitting, define the pseudo-outcome $\psi = (Y - \mu(X))(W - e(X))$ and $\tau = \arg \min_f \mathbb{E}[(\psi - f(X))^2]$. The PO-risk statistic $\mathbb{E}[\tau(X)^2]$ (and the R -risk analogue) is large under concept drift and *insensitive to pure covariate shift*; its permute-then-refit p -value gives the concept-drift test. Thus MMD and PO-risk are complementary detectors, not substitutes.

3 Localization: attribution by ablation and by coalitions

Write $\widehat{\text{MMD}}_u^2(S)$ for the statistic computed on the sub-vector of features indexed by a subset S of the groups $\mathcal{G}$ (fixed kernel; only the *ranking* of importances is used, so scale is immaterial).

Definition 4 (LOGO-MMD). *The leave-one-group-out importance of group g is $\phi_g^{\text{LOGO}} = \widehat{\text{MMD}}_u^2(\mathcal{G}) - \widehat{\text{MMD}}_u^2(\mathcal{G} \setminus \{g\})$.*

LOGO is the single marginal contribution *at the full coalition*. Its blind spot is redundancy: two duplicated shifted groups each obtain $\phi^{\text{LOGO}} \approx 0$.

Definition 5 (Shapley-MMD). *Treating $v(S) = \widehat{\text{MMD}}_u^2(S)$ as a coalitional game, the Shapley value of group g is*

$$\phi_g^{\text{Sh}} = \sum_{S \subseteq \mathcal{G} \setminus \{g\}} \frac{|S|! (K - |S| - 1)!}{K!} (v(S \cup \{g\}) - v(S)), \quad K = |\mathcal{G}|. \quad (3)$$

Proposition 2 (Efficiency and exact enumeration). $\sum_g \phi_g^{\text{Sh}} = v(\mathcal{G}) - v(\emptyset) = \widehat{\text{MMD}}_u^2(\mathcal{G})$, so the Shapley values additively decompose the total divergence. With K groups there are 2^K subsets; for the small K here ($K \approx 7$) the game is enumerated exactly, requiring no approximation and no gatekeeping.

Standalone MMD $v(\{g\})$ (marginal at $\emptyset$), LOGO (marginal at $\mathcal{G}$), and Shapley (average over all S) form a spectrum; unlike LOGO, MMD^2 is not additive across groups, which is why Shapley—satisfying efficiency—is the principled decomposition and LOGO the cheap ranking.

Remark 1 (LOGO $\equiv$ LOCO in a different metric). *The concept-drift analogue replaces the divergence v by a causal risk: leave-one-covariate-out (LOCO) importance $\rho_g = \text{risk}(\mathcal{G} \setminus \{g\}) - \text{risk}(\mathcal{G})$ with the PO-/R-risk. The same ablation principle instantiated on MMD (marginal, covariate) and on risk (conditional, outcome) covers the two shift types.*

4 Two-stage selection by permutation and stability

We deliberately avoid the Benjamini–Hochberg (BH) family. In this problem the number of group hypotheses is small ($K \approx 7$) and the group statistics are strongly correlated (aggregations of the same raw attribute, and correlated raw attributes); BH’s independence/PRDS assumptions are dubious and, with so few hypotheses, BH is simultaneously under-powered and poorly calibrated in practice. We use permutation-based control instead, which is exact under exchangeability and correlation-robust.

Stage 1 (groups). Rank groups by Shapley-MMD; select $\mathcal{G}^* = \{g : p_g < \alpha_1 \text{ and } \hat{\pi}_g \geq \pi_{\text{thr}}\}$, where p_g is the group’s permutation p -value and $\hat{\pi}_g$ its subsampling selection frequency (Section 5). Combining a permutation test with a stability threshold is both practical and finite-sample controlled.

Stage 2 (features within a selected group). Within each $g \in \mathcal{G}^*$ apply a *Westfall–Young step-down max-T* permutation procedure. Let $T_j = \widehat{\text{MMD}}_u^2(\{j\})$ for feature $j \in g$. Draw B label permutations *shared across the features of g* ; from each permutation b form the maximum $M_b = \max_{j \in g} T_j^{(b)}$. The adjusted p -value is

$$\tilde{p}_j = \frac{1 + \sum_{b=1}^B \mathbb{1}\{M_b \geq T_j\}}{1 + B}, \quad (4)$$

and feature j is selected if $\tilde{p}_j \leq \alpha_2$.

Proposition 3 (Exact FWER control). *Under the family null and exchangeability, using the shared-permutation max statistic gives $\mathbb{P}(\exists j \text{ null} : \tilde{p}_j \leq \alpha_2) \leq \alpha_2$, exactly and without independence assumptions; because the null is built from the same permutations for all features, the procedure adapts to their correlation.*

The same construction applies at every split of the multi-resolution tree root $\rightarrow$ attribute $\rightarrow$ aggregation family $\rightarrow$ feature: each node’s children are tested by shared-permutation max-T, and the tree is *gated*—a node is entered only if its parent was selected—so the family-wise guarantees compose down a hierarchical, correlation-robust, BH-free procedure. Each node is re-tested on its own feature subset (marginal at the root, increasingly conditional deeper down): *subset post-hoc localization* conditional on a significant global test.

5 Stability selection and the MMD bootstrap

To quantify selection stability we resample and record the selection frequency $\hat{\pi}_g$ and a percentile interval of ϕ_g ; a group is stable if $\hat{\pi}_g \geq \pi_{\text{thr}}$ and its interval excludes 0.

Remark 2 (Bootstrap ties bias the U-statistic). *With-replacement resampling produces distinct positions $i \neq j$ mapping to the same original point, for which $k(x_i, x_j) = 1$ enters the off-diagonal sums in (1). In a resample of size m the expected number of such collisions is $\approx m-1$, inflating each within-sample term by $\approx (1-c)/(m-1)$ while the cross term is unaffected; hence $\widehat{\text{MMD}}_u^2$ is biased upward, and the effect dominates near H_0 where the true value is ≈ 0 . Remedies: additive jitter (a form of kernel smoothing / randomized smoothing), m -out-of- n subsampling without replacement (no ties), generalized diagonal exclusion of same-origin pairs, or the wild bootstrap for degenerate kernel statistics. We use the permutation test for significance (no ties) and the bootstrap only for stability of already-significant groups, where these effects are second order.*

6 Two-dimensional structure

Running the pipeline on two axes (merchant and buyer) yields two feature libraries from the *same* order table. If the batch labels $W_{\text{merchant}}, W_{\text{buyer}}$ are (near-)independent and the injected shifts act on disjoint raw attributes, aggregating along one axis averages out the other axis’s shift, so each axis recovers only its own drivers. This separates a *supply-side* change from a *demand-side* change.

7 Characterization and the retrain decision

7.1 Per-feature characterization

For a selected feature we report the standardized effect $d = (\bar{x}_1 - \bar{x}_0)/s_{\text{pool}}$, per-batch quantile grids, and the KS-optimal split $t^* = \arg \max_t |\hat{F}_1(t) - \hat{F}_0(t)|$ with its KS statistic and per-batch mass below t^* ; t^* materializes into an executable `candidate_rule`.

7.2 Impact decomposition

Let a downstream model $\hat{f}$ be trained on the old batch with per-point loss ℓ . Define the in-distribution loss $L_{\text{old}} = \mathbb{E}_{P_0}[\ell]$, the new loss $L_{\text{new}} = \mathbb{E}_{P_1}[\ell]$, and the importance-weighted estimate

$$L_{\text{iw}} = \mathbb{E}_{P_0}[w(x)\ell(x)], \quad w(x) = \frac{p_1(x)}{p_0(x)} = \frac{e(x)}{1 - e(x)} \cdot \frac{\mathbb{P}(W = 0)}{\mathbb{P}(W = 1)}, \quad (5)$$

using the propensity e as the density ratio.

Proposition 4 (Covariate/concept decomposition). *If $P(Y \mid X)$ is unchanged (pure covariate shift) and the supports overlap, then $L_{\text{iw}} = L_{\text{new}}$, so covariate impact $:= L_{\text{iw}} - L_{\text{old}}$ captures the degradation and concept impact $:= L_{\text{new}} - L_{\text{iw}} = 0$. A significant concept impact therefore indicates concept drift.*

7.3 Overlap gate

The identity requires overlap. Two diagnostics: the weight effective sample size $\text{ESS} = (\sum w)^2 / \sum w^2$, and—robust to weight clipping—the propensity AUC (batch separability). When $\text{AUC} \approx 1$ the batches are (near-)separable, w is ill-defined, and the covariate/concept split is not identifiable label-free.

7.4 Decision rule

$$\text{decision} = \begin{cases} \text{retrain / collect labels,} & \text{AUC} > \tau_{\text{AUC}} \text{ and } (L_{\text{new}} - L_{\text{old}})/L_{\text{old}} > \delta, \\ \text{retrain,} & \text{concept-impact CI} > 0, \\ \text{reweight,} & (L_{\text{iw}} - L_{\text{old}})/L_{\text{old}} > \delta, \\ \text{monitor,} & \text{otherwise.} \end{cases} \quad (6)$$

Aggregation typically makes a detectable shift separable ($\text{AUC} \rightarrow 1$), so at the entity level the honest verdict is usually “retrain / collect labels,” with localization indicating *what* to label.

8 Structured insight contract

Each finding is emitted as a typed object (`fsds-insight/1.0`): identity (`schema_version`, `insight_id`=`hash(dedup_key=axis:scope:target, generated_at)`); a three-axis confidence {significance p , stability $\hat{\pi}$, identifiability Δ }; a `trend` (`status` $\in$ {new, worsening, improving, stable, resolved}, `priority`, `first_seen`) obtained by diffing against a previous run keyed on `dedup_key`; per-feature `candidate_rule`; and routing/ranking (`owner`, `priority` = `exposure` $\times$ `impact`, `decision`). The stable identity supports upsert/dedup; the same JSON drives dashboards, alert routing, and feature-store governance.

9 Pipeline

1. Aggregate orders to the entity axis: $x_j = \Phi(O_j)$.
2. Global detection: MMD permutation test (covariate); PO-/R-risk (concept).
3. Stage 1: Shapley/LOGO ranking; group permutation p_g and stability $\hat{\pi}_g \rightarrow \mathcal{G}^* = \{p_g < \alpha_1, \hat{\pi}_g \geq \pi_{\text{thr}}\}$.
4. Stage 2: within each $g \in \mathcal{G}^*$, Westfall–Young max-T permutation at $\alpha_2 \rightarrow$ selected features.
5. Characterize each selected feature (d , quantiles, KS split, rule).
6. Decision: train downstream model on old batch; compute $L_{\text{old}}, L_{\text{iw}}, L_{\text{new}}$, ESS, AUC $\rightarrow$ decision.
7. Emit `fsds-insight/1.0` objects (two-level: group headline + feature drill-down), diff against previous run for `trend`.

10 Synthetic validation

On a synthetic marketplace (250 merchants, 800 buyers, 20,000 orders, 69 features/axis) with covariate shift injected on {`gmw`, `user_rating`} (merchant axis) and {`basket_size`, `discount_rate`} (buyer axis):

Axis	Global p	Recovered drivers	Leaf false discoveries
merchant (W_{merchant})	0.003	{ <code>gmw</code> , <code>user_rating</code> }	0
buyer (W_{buyer})	0.003	{ <code>basket_size</code> , <code>discount_rate</code> }	0

Exact Shapley-MMD ranks the true drivers top-2 on each axis (permutation $p = 0.0099$, bootstrap top-2 frequency 1.00). The decision self-check (with support overlap) yields *reweight* for covariate-only shift (AUC = 0.77), *retrain* for concept drift (AUC = 0.51), and *monitor* for a shift on unused features (AUC = 0.92).

11 Limitations

- MMD is $O(N^2)$; large axes need linear-time MMD or subsampling.
- Aggregation tends to make detectable shifts separable, so label-free covariate/concept attribution is often infeasible at the entity level.

- Concept drift requires (possibly delayed) labels.
- Permutation validity assumes exchangeability; correlated entities need block permutation. Reported post-selection intervals should use FCR correction.

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